End Semester Examination 2024-25

Name of the Program: **M. Tech (CSE)**
Name of the Course: **Advance Operating Systems**
Time: **3 Hours**

Semester: ***I***
Course Code: **MCS-103**
Maximum Marks: **100**

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1 **(20 marks)**

a) **Consider the following table (Here lower number in priority column represents higher priority of process):**

Process	Arrival Time	Burst Time	Priority
1	03	06	3
2	07	10	1
3	06	08	4
4	02	12	2

For following scheduling algorithms, determine the turnaround time and waiting time for each process and the average turnaround time for all jobs.
a) Priority (Preemptive).
b) Round-robin with a time quantum of 2 unit.

b) **What are major design types for operating system kernels? Which of these types better satisfies the requirement "Convenient access to operating system data structures"? Justify your answers.**

c) **What are the necessary conditions for a deadlock to occur in a system? Explain how the Banker's Algorithm can be used to avoid deadlocks.**

CO1

Q2 **(20 marks)**

a) **What are necessary conditions to solve critical section problems? Explain how Peterson solution satisfies various conditions to solve critical section problems.**

b) **Describe seven state process transition diagram? Provide significance of each state.**

c) **Explain semaphore. Compare and contrast the performance of busy-waiting and without busy waiting semaphores in different scenarios.**

CO2

Q3 **(20 marks)**

a) **Compare and contrast Contiguous and Non Contiguous memory allocation schemes by taking suitable example.**

b) **How many page-faults would occur for the following reference string, for three page frame using LRU and Optimal algorithms:**
1 , 2 , 3, 4, 5, 5, 3, 2, 4, 7 , 8 , 7 , 3 , 9 , 7 , 6, 4, 5, 4, 5, 4, 2

c) **Consider a single level paging scheme. The virtual address space is 256 MB and page table entry size is 4 bytes. What is the minimum page size possible such that the entire page table fits well in one page?**

CO3

Q4 **(20 marks)**

a) **Discuss the technical challenges and potential solutions associated with implementing process migration in a large-scale distributed system.**

CO4

b)	How dead lock is detected in Distributed environment. Explain various Distributed Deadlock Detection Strategies	
c)	Explain distributed snapshot algorithm by taking suitable example.	
Q5	(20 marks)	
a)	Suppose that a disk drive has 500 cylinders, numbered 0 to 499. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 78, 147, 91, 174, 194, 250, 102, 475, 330 Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for SCAN and LOOK disk-scheduling algorithms?	CO5
b)	Explain process management in LINUX and WINDOWS systems.	
c)	A paging scheme uses a Translation Lookaside buffer (TLB). The effective memory access takes 160 ns and a main memory access takes 100 ns. What is the TLB access time (in ns) if the TLB hit ratio is 60% and there is no page fault?	